

DATABASE SYSTEMS COURSE PROJECT PROPOSAL

Section: BCS-4H

Spring 2026

Project Title: Advanced E-Commerce System

Team Members:

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Introduction and Brief Description of the System:

The Advanced E-Commerce System is a web-based application designed to facilitate online buying and selling of products. The system allows customers to browse products, add items to their cart, place orders, and make payments securely. It also enables administrators to manage products, categories, users, and monitor overall system activity.

The purpose of this system is to provide a complete digital shopping experience while maintaining efficient data management using a relational database. The system solves problems related to manual inventory handling, inefficient order processing, and lack of centralized data management.

System Functionality

The system will perform following major functions:

- User registration and login
- Product browsing and searching
- Category-based product filtering
- Add to cart and manage cart
- Place orders and checkout
- Online payment processing
- Order tracking and history
- Product reviews and ratings
- Admin product management (Add, Update, Delete)
- Inventory/stock management
- Sales and order reports generation

Users of the System:

The system will have the following user roles:

1. Administrator

- Manage products (add/update/delete)
- Manage categories
- View and manage all orders
- Monitor users and system activity
- Generate reports (sales, inventory)

2. Customer

- Register and login
- Browse and search products
- Add/remove items from cart
- Place orders and make payments
- View order history
- Give reviews and ratings

3. Seller (Optional)

- Add and manage their own products
- View their product sales
- Update product details and stock

Technologies Used:

The system will be implemented as a web application using the following technologies:

Front-End:

- HTML
- CSS
- JavaScript
- Bootstrap

Back-End:

- Node.js (maybe changed)

Database:

- MySQL

Scope of the System:

System will Cover:

- Complete user management system
- Product and category management
- Shopping cart functionality
- Order and payment processing
- Review and rating system
- Database operations (CRUD)
- SQL queries with joins, views, and transactions

System will not Cover:

- Real-time payment gateway integration (simulation only)
- Mobile application version
- Advanced AI-based recommendation system
- Third-party logistics integration

Assumptions:

- All users have basic knowledge of using a web application
- Payments will be simulated (no real financial transactions)
- Each product belongs to one category
- Stock is managed centrally by the system
- Users will provide valid and accurate information